Inheritance, Variation and Evolution

Complete some of the tasks from the grid below to reach a total of _____ points over this unit of work. Try and cover a variety of tasks over the unit so that you're practising different skills. Once you've completed a task, colour that box on the grid to keep a record of your points. Can you get the highest point score this unit?

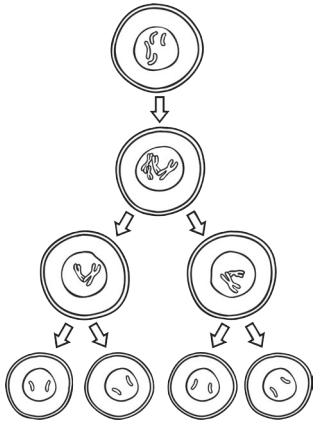
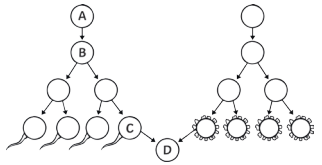
Topic	1 Point	2 Points	4 Points	6 Points	10 Points
Reproduction and Meiosis	State what mitosis and meiosis are used for.	Draw a diagram to show how meiosis occurs.	Compare sexual and asexual reproduction.	Pick an animal and find out how many chromosomes are contained in each of its cells. Draw a diagram that shows how gametes are formed and fertilisation occurs. Label the number of chromosomes in each cell type.	Produce a video that explains the difference between mitosis and meiosis.
DNA and the Genome	Describe where DNA is found.	State one reason why it is important to study the human genome.	Make a model that shows where DNA is found in a cell, using whatever materials you'd like.	Write a newspaper article to persuade people that it is important to invest money in studying the human genome.	Research the scientist Rosalind Franklin. Explain why many people are dissapointed that she didn't receive a Nobel prize for her work.
Inheritance	Describe the difference between genotype and phenotype.	Draw a diagram to show the relationship between genes and chromosomes.	Create a revision resource that will help you to learn the keywords for this topic.	If the **d** allele results in dimples and the **D** allele does not, draw a genetic diagram to show the inheritance of dimples in	Find out how Huntington disease is inherited. Produce a genetic diagram to explain the probability of a

				the children of a woman with dimples and a man who is heterozygous for the dimples allele.	heterozygous dad with the disease and a mum without the disease, having children who inherit it.
Genetic Disorders	State how cystic fibrosis and polydactyly are inherited.	Explain how your sex was determined.	Give two reasons for and two reasons against embryo screening.	Write an information guide for parents who are considering embryo screening that tells them the benefits and risks associated with the procedure.	A couple are undergoing IVF treatment. They already have a child with sickle cell anaemia which is a painful disorder that shows recessive inheritance. Evaluate the use of embryo screening for this couple.
Variation	Draw a Venn diagram for genetic and environmental variation. Fill it with examples of variations between individual humans.	Explain what causes new variations to arise.	Produce a Facebook profile page for Darwin whilst he's on his voyage. Which locations does he check in to? What does he post about, and who comments?	Research some other species that Darwin recorded on his voyage. What genetic and environmental variations might he have found between individuals of the species?	Collect data on a variable (e.g. height, eye colour) from your friends and family. Plot your data on an appropriate graph.
Evolution	What is meant by the term 'natural selection'?	Write definitions for the keywords used during this lesson.	Use the development of Darwin's theory to explain how scientific theories develop over time.	Use Darwin's theory of natural selection to explain how the elephant's trunk evolved.	Produce a comic strip that shows how Darwin's theory of evolution by natural selection works.

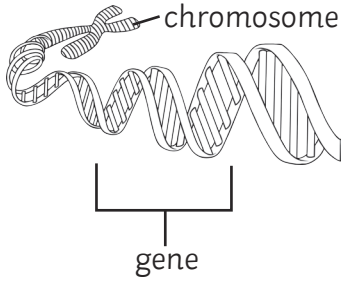
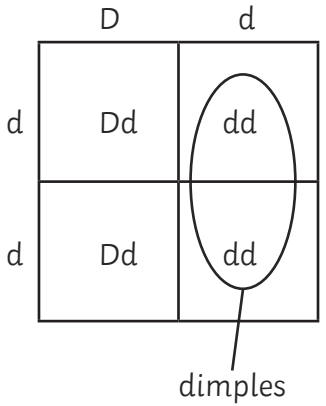
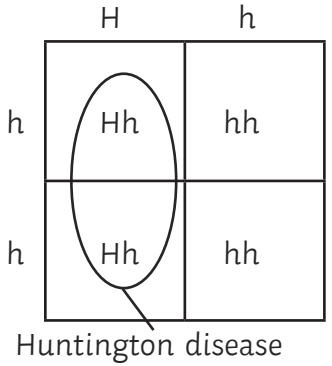
Evidence For Evolution: Fossils	What can we learn from fossils?	Explain why scientists can't be certain how life began.	Use your knowledge from physics to explain how scientists are able to date fossils.	Produce a poster aimed at primary school children that explains the different ways in which fossils might form.	Research the fossil record of a different organism to the ones we studied in class. Explain how the species has changed over time and what might have contributed to these changes.
Evidence For Evolution: Resistant Bacteria	What caused the rise in MRSA?	Write a patient leaflet to be displayed in doctors' waiting rooms. It should explain why doctors might not prescribe antibiotics and what patients should do if they are given a course of antibiotics.	Produce a storyboard that shows how bacteria develop antibiotic resistance.	Penicillin is an antibiotic. It was used a lot in the past to treat infections but there are now penicillin-resistant strains of bacteria. Explain how natural selection could have produced these strains of penicillin-resistant bacteria.	Produce a video or animation that shows how antibiotic resistance develops in bacteria.
Selective Breeding	What is selective breeding?	Describe the steps involved in selective breeding.	Write a social media post about the positive or negative side of selective breeding. Then add a comment on the post that provides the opposite view.	Conduct your own research into selective breeding. Find one example of selective breeding for good, and one example of selective breeding gone wrong. Explain the case in your own words, highlighting the advantages/	Make a video or animation that shows the process of selective breeding.

				disadvantages. You can present this information in any way you choose.	
Genetic Engineering	What is genetic engineering?	Give some uses of genetic engineering.	Research a use of genetic engineering that was not covered in the lesson.	Produce an information leaflet that informs the public how genetic engineering is used in agriculture.	Make a film or write a script that describes the advantages and disadvantages of genetic engineering.
Classification	Describe how organisms are named and give an example.	Describe how Linnaeus classified living things and give an example.	Draw a diagram that shows the three-domain system of classification, including the kingdoms in each. Give an example of an organism that would fit in each kingdom and explain why.	Produce a presentation that explains how evolutionary trees are used by scientists to show how organisms are related. Include the different information that could be used to produce a tree and how this might affect the shape of the tree. Explain what kind of data is used to place extinct animals into a tree. Include examples of some relationships.	Use your knowledge from the whole unit to explain, with examples, how scientific methods and theories develop over time.

Inheritance, Variation and Evolution Answers

Topic	1 Point	2 Points	4 Points	6 Points	10 Points
Reproduction and Meiosis	**Mitosis is used for growth and replacement of cells. Meiosis is used to produce gametes.**		**Sexual reproduction involves two parents, whilst asexual reproduction only needs one. Sexual reproduction involves gametes which are produced by meiosis. There is no fusion of gametes in asexual reproduction. Only mitosis is involved in asexual reproduction. Asexual reproduction produces genetically identical offspring (clones). However, there is mixing of genetic information in sexual reproduction that causes variation in the offspring.**	**The chromosome numbers will depend on the animal the student has chosen, but you're looking for them correctly identifying which cells will have half the number of chromosomes. The diagram should look similar to the following.** 	**Meiosis occurs in the ovary or testes and produces gametes for sexual reproduction. Mitosis occurs in all other cells - it is asexual reproduction that is used for growth and replacement of cells. Meiosis forms 4 daughter cells, in two divisions, and each has 23 chromosomes. Mitosis forms two daughter cells, in one division, and each has 46 chromosomes. Genetic information is not mixed in mitosis, so daughter cells are identical. In meiosis, there is mixing of the genetic information that causes variation in the daughter cells.**

DNA and the Genome	**DNA is found in chromosomes in the nucleus.**	**One of:** **It helps us to search for genes that are linked to different types of disease.** **It helps us to understand and treat inherited disorders.** **We can use the information to trace human migration patterns from the past.**	**This will vary, depending on students choice of materials, but they should have found a way of showing that DNA is found in chromosomes and these are found in the nucleus.**	**The article should include the following points:** **It helps us to search for genes that are linked to different types of disease.** **It helps us to understand and treat inherited disorders.** **We can use the information to trace human migration patterns from the past.** **The best artices will use persuasive language and may give examples to engage the reader, such as examples of inherited diseases and the impact they can have.**	**Many people can name Watson and Crick as the scientists who modelled the structure of DNA, however they are less likely to have heard of Rosalind Franklin. It was her work using x-rays to study DNA that led Watson and Crick to complete their model. Many people felt she should have shared the Nobel prize for the discovery. Students may also refer to the pioneering work that she did on the tobacco mosaic virus and polio.**

Inheritance	**The genotype is the combination of alleles that someone inherits and the phenotype is the trait that is expressed as a result of the genotype.**		**Students should have included the keywords: gene, allele, heterozygous, homozygous, dominant, recessive, genotype and phenotype.** **The best examples will include a creative way to record the keywords and definitions.**		**Huntington disease shows dominant inheritance.** **Male genotype: Hh.** **Female genotype: hh.** 
Genetic Disorders	**Cystic fibrosis inheritance is recessive. Polydactyly inheritance is dominant.**	**The student received an X chromosome from their mum. If they are female they received an X chromosome from their dad and if they are male they received a Y chromosome from their dad.**	**A list of reasons for and against is provided in lesson 4, Genetic Disorders.**	**Students may pick different issues to cover but the information should be balanced. The guide should definitely include the risk of miscarriage and the benefits of being able to make an informed decision. The best leaflets will discuss the emotional impact of testing on the family.**	**Only healthy embryos would be inserted into the womb. This way, the parents would have a high chance of preventing their next child having the disorder.** **Spare embryos would be destroyed, some people disagree with destroying this potential life.**

					It would reduce the incidence of the disease. The procedure is expensive, but it would save long-term health costs of a child born with the disorder.
Variation	Variation such as scars and tattoos in the environment section. Variation like eye colour and tongue-rolling is genetic. It could be argued that the majority of variations are affected by both, e.g. natural skin colour is genetic, but affected by sun exposure.	Random mutations in the DNA can cause a change in a species over a long time, if the new phenotype is useful in the environment.	The best examples would show Darwin sharing examples of species he has discovered and why he thought they were interesting. Students might have him check in at the location those species were found, or have other people ask questions about his discoveries.	Darwin found many species on his travels, including some large fossils and many different types of insects. Students should suggest sensible variations that would be present between individuals and decide if they are genetic, environmental or both.	Continuous data should be plotted on a scatter graph while discrete data should be plotted on a bar chart.

Evolution	**Natural Selection: the process by which the individuals that are more able to compete can survive to reproduce.**	**Variation: differences in traits between individuals of a population.** **Competition: individuals in a population attempt to gain a share in the available resources.** **Adaptations: a trait that is better suited to the environment.** **Natural Selection: the process by which the individuals that are more able to compete can survive to reproduce.** **Inheritance: Alleles are passed on from parent to offspring.**	**Darwin collected a lot of evidence that helped him form and adapt his theory of evolution. He collaborated with many other scientists who gave him further evidence and ideas to develop his theory into the one he eventually published.**	**There was variation in trunk length. The elephants competed for food and water, and those with the longer trunks were better able to compete. These elephants were more likely to survive and reproduce - passing the alleles for their long trunk to their offspring.**	**The best comics would include: variation between individuals in a trait, competition for a resource, some of the individuals surviving whilst others don't, reproduction and offspring carrying the same trait as the parents.**

Evidence For Evolution: Fossils	**We can learn how much or how little different organisms have changed as life has developed on Earth.**	**Earliest life forms were soft-bodied so they decayed. The conditions needed for fossils to form are rare. Geological activity has destroyed a lot of fossils. We haven't discovered all the fossils yet.**	**Scientists measure radioactive decay from the rocks around the fossils. As they know the rate at which certain elements decay, they are able to determine how old the rocks are.**	**Students should have clear, simple explanations with diagrams that show:** **1) conditions for decay are absent (amber, peat, ice);** **2) minerals replace the hard parts of the organism;** **3) traces of organisms are preserved.**	**This task will vary, depending on the organism that the student picks. They should have identified changes in the fossil record that show how the species has evolved, and suggestions about why they might have evolved in this way.**

Evidence For Evolution: Resistant Bacteria	**Doctors overused antibiotics. Patients didn't finish their courses of antibiotics. There was an overuse of antibiotics in agriculture.**	**Your doctor will not prescribe antibiotics if you have a virus because antibiotics only work on bacterial infections. Your doctor will not prescribe antibiotics if your infection is not serious and he/she thinks your body will be able to fight the infection itself. If you are given antibiotics, it's very important you complete the course so that there is less opportunity for resistant strains of bacteria to develop.**	**A storyboard that shows multiple bacteria with a random mutation in one bacterium that gives it antibiotic resistance. The use of an antibiotic that kills the other strain but that the resistant strain is more able to compete, survive and multiply. A further random mutation may give resistance to another antibiotic.**	**A random mutation in the bacteria gave resistance to penicillin. Therefore, some bacteria survive the penicillin and these reproduce by binary fission. The gene for penicillin resistance is passed on to the offspring.**	**An animation or video that shows multiple bacteria. A random mutation in one bacterium that gives antibiotic resistance. The use of an antibiotic that kills the other strain but that the resistant strain is more able to compete, survive and multiply. A further random mutation may give resistance to another antibiotic.**

Selective Breeding	**Humans control selection by choosing organisms to breed that have desired traits.**	**1. Choose the plants or animals from your existing stock that have the feature you want.** **2. Breed these with each other.** **3. Choose the best of the offspring and breed these together.** **4. Continue this process over several generations.** **5. Eventually, all the offspring have the features that you are looking for.**	**This will vary, depending on the student's social media post and the choice of argument. However, you're looking for an argument with supporting evidence either for or against selective breeding, with a comment to balance the argument.**	**This will depend on the example that the student finds. You're looking for a balanced argument that gives both advantages and disadvantages of selective breeding, backed up by evidence from their research. This shouldn't be an example they studied in class.**	**You're looking for human involvement in selection - breeding of two organisms with the desired trait and then selection of the offspring with examples of this trait. The student should have shown that this repeats over many generations.**
Genetic Engineering	**Genetic engineering is a process that involves modifying the genome of an organism by introducing a gene from another organism to give a desired characteristic.**	**Golden rice to reduce vitamin A deficiency. Producing insulin for diabetics. GM crops that produce a larger yield or grow in more difficult climates. Animal models of disease.**	**This will depend on what the students research.**	**The leaflet should cover information about GM crops and the advantages of engineering them, as well as the concerns people may have.**	**Possible advantages:** **• GM crops have bigger yields and therefore have increased food value. This may help us to address an expanding population.**

					• Animals and plants grow faster. • We can mass produce human proteins. • GM crops can grow in dry, hot, cold or flooded parts of the world. This may help us to manage the effects of global warming on agriculture. • GM crops can produce their own pesticide, can be resistant to herbicides, or can contain specific nutrients. • Models of human diseases can help us to develop treatments.

					Possible disadvantages: • Nobody can be sure of the long-term effects of eating GM food on human health. • Pesticide-producing crops may result in insects developing resistance to pesticides. • Genes from GM crops or animals could spread to species in the wild. • People worry that this may lead to the manipulation of human genes to make 'designer' children.

Classification	**Binomial naming system of genus and then species, with an example of a binomial name for an organism.**	**Kingdom, phylum, class, order, family, genus, species and an example of an organism and which of each it belongs to.**	**The diagram should include: archaea, with the kingdom archaeabacteria; bacteria, with the kingdom eubacteria; and eukaryota with the four kingdoms: protists, fungi, plants and animals.** **Students may use an explanation like, 'the cells don't have a nucleus and they are found in extreme conditions'.**	**This should explain how morphology can be compared to look for similarities between organisms, but that DNA and protein sequencing is more accurate and can show different relationships. Fossil data will be used to show where extinct animals may have fitted into an evoutionary tree. The best examples will include illustrations of evolutionary trees and information about relationships that students have researched.**	**Students could include information about microscopes and DNA sequencing changing the way in which organims are classified. Microscope methods allowing scientists to see chromosomes in cells and observe their behaviour in mitosis and meiosis, therefore confirming Mendel's ideas on units of inheritance. X-rays being used to show the structure of DNA. More data on speciation being used to support Darwin's theory of evolution by natural selection.**